

Mean sudoku

Answer: 368974251
615249378 254197863

Solution: We refer to the nine 3×3 boxes of the Sudoku grid as B_1 through B_9 , numbered in standard reading order (left to right, top to bottom).

First, let us consider the requirement that the sums of the circled numbers in each box result in the values 1 through 9. There is only one way to represent 1 and 2 as a sum of distinct digits, namely $\{1\}$ and $\{2\}$. Since there are exactly two boxes containing a single circled cell, all other target sums must be represented by at least two distinct digits. This gives the unique partitions:

$$3 \mapsto \{1, 2\}, 4 \mapsto \{1, 3\}.$$

For the sum of 5, we have two possibilities:

$$5 \mapsto \{1, 4\} \text{ or } \{2, 3\}.$$

There are exactly three boxes containing two circled cells (corresponding to the sums 3, 4, and 5). Consequently, the remaining target sums (6 through 9) must be formed by three distinct digits.

3	4	5	6	7	8	2	9	①
6	②	9	①	③	④	5	7	8
8	7	1	5	9	2	4	6	③
9	③	6	2	8	7	1	5	4
7	⑤	8	4	①	6	9	③	②
4	①	2	9	5	3	7	8	6
②	9	4	3	6	5	8	①	7
5	8	③	7	4	①	6	②	9
①	6	7	8	②	9	3	④	5

This yields the following combinations:

$$6 \mapsto \{1, 2, 3\}, 7 \mapsto \{1, 2, 4\},$$

$$8 \mapsto \{1, 3, 4\} \text{ or } \{1, 2, 5\},$$

$$9 \mapsto \{1, 2, 6\}, \{1, 3, 5\}, \text{ or } \{2, 3, 4\}.$$

Now, consider the single circled cell in box B_1 . Its value is restricted to 1 or 2. By Sudoku rules, this value cannot appear in any circled cell in the adjacent boxes B_2 or B_4 . Since there is only one possible triplet representation that does not contain the digit 1, we conclude that the circled cell in B_1 must be 2. Consequently, the circled triplet in box B_2 is either $\{1, 3, 4\}$ or $\{1, 3, 5\}$, and similarly, the circled triplet in box B_4 is either $\{1, 3, 4\}$ or $\{1, 3, 5\}$.

It follows that the single circled cell in box B_5 is 1. Therefore, the circled pair in box B_6 must be $\{2, 3\}$, since all other possible pair representations contain the digit 1. The circled triplet in box B_9 cannot contain both 2 and 3, so it must be $\{1, 2, 4\}$. This forces the circled pair in B_6 to be arranged with 3 in the middle cell and 2 in the side cell. Consequently, the circled pair in B_3 is determined to be $\{1, 3\}$. Finally, in box B_7 the circled triplet is $\{1, 2, 3\}$, and in box B_8 the circled pair is $\{1, 2\}$.

Placing these values into the grid in a way that respects the standard Sudoku constraints (rows, columns, and boxes) yields the following layout:

			6	7				(13)
6	(2)		(1345)	(1345)	(1345)		7	
	7						6	(13)
	(1345)	6			7			
7	(1345)			(1)	6		(3)	(2)
	(1345)					7		6
(123)				6			(124)	7
		(123)	7		(12)	6	(124)	
(123)	6	7		(12)			(124)	

→

			6	7				(13)
6	(2)		(1345)	(345)	(1345)		7	
	7						6	(13)
	(1345)	6			7			
7	(45)			(1)	6		(3)	(2)
	(1345)					7		6
(123)				6			(124)	7
		(23)	7		(1)	6	(24)	
(13)	6	7		(2)			(14)	

Next, we consider the yellow cells. Observe that if a given box contains n yellow cells, the sum of the digits in those cells must be divisible by n so that their average is an integer.

In box B_4 , there are 8 yellow cells. Let x denote the digit in the single colourless cell. Since the sum of the digits 1 through 9 is 45, we require $45 - x$ to be divisible by 8. This implies $x = 5$. Consequently, the average of the yellow cells in box B_4 is 5. In box B_3 , let x and y be the digits in the two colourless cells. Then $45 - (x + y)$ must be divisible by 7 (since there are 7 yellow cells), so the sum $x + y$ must be an element of the set $\{3, 10, 17\}$. The case $x + y = 10$ is impossible because it would result in an average of 5, which is already assigned to box B_4 . From our earlier deductions regarding the circles, we know that one colourless cell $y \in \{1, 3\}$. This constraint forces $x + y = 3$; hence, the colourless cells are $\{1, 2\}$. This yields an average of 6 for box B_3 . Similarly, in box B_8 , the colourless cells are determined to be $\{8, 9\}$. The sum of the yellow cells is therefore $45 - 17 = 28$, resulting in an average of 4 for box B_8 .

In box B_7 , let x denote the digit in the colourless cell without a circle. The sum $4 + 5 + 6 + 7 + 8 + 9 - x$ must be divisible by 5, which implies $x = 4$. Placing this digit gives an average of 7 for box B_7 .

There is only one way to obtain averages 1 and 9, namely from the singleton sets $\{1\}$ and $\{9\}$. Since there are exactly two boxes with a single yellow cell, this implies that the yellow cell in box B_9 is 1, and the yellow cell in box B_5 is 9.

We are left with three boxes - B_1 , B_2 , and B_6 - and three remaining averages: 2, 3, and 8. The average 2 can only occur in box B_2 , which forces its yellow cells to be $\{1, 3\}$. The average 8 cannot occur in box B_5 , so it must occur in box B_1 , where the remaining yellow cells are f8, 9g. Finally, box B_5 has an average of 3, so its remaining yellow cells are $\{1, 4, 5\}$.

After placing these digits and simplifying to satisfy all Sudoku constraints (note that the right-middle cell in B_2 must be 4 because the circled triplet in B_4 is $\{1, 3, 5\}$), we obtain:

			6	7		2		①
6	②	89	⑬	⑬	⑬④⑤		7	
89	7						6	⑬
	⑬④⑤	6			7	⑭⑤	⑭⑤	⑭⑤
7	⑤			①	6		③	②
	⑬④⑤		9			7		6
⑫③		4		6			①	7
		⑫③	7		①	6	⑫④	
⑬	6	7	89	②	89		⑭④	

→

			6	7		2		①
6	②	89	①	③	④		7	
89	7						6	③
	③	6			7	1	5	4
7	⑤			①	6		③	②
	①		9			7		6
②		4		6			①	7
		③	7		①	6	②	
①	6	7	8	②	9		④	

Finally, we fill the grid by verifying the basic Sudoku rules. One possible approach is to solve box by box, for example in the order: B_6 , B_9 , B_7 , B_4 , B_1 , B_5 , B_2 , B_3 , and B_8 .